

CHAPTER IV: Tensor Star Decomposition & Multilinear S13 Discrete Manifolds[

HOSVD, Balanced-Ternary Core Contraction, and SO(5) Geodesic Isomorphism

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Status: Chapter IV of *Little Nistam and The Lattice of Harmony*

Target Engine: crates/gemma-s13/src/star_tensor.rs (11/11 Verified Tests)

Executive Summary

This chapter establishes the multilinear algebraic foundation and physical silicon implementation of **Tensor Star Decomposition** combined with **S13 Balanced-Ternary Core Quantization**. An N-way continuous tensor factorizes into a central dense discrete balanced-ternary core G connected to peripheral continuous factor leaves $U^{\wedge}(n)$: $X \sim G \otimes x_1 \otimes U^{\wedge}(1) \otimes x_2 \otimes U^{\wedge}(2) \dots \otimes x_N \otimes U^{\wedge}(N)$.

Key proven properties include: (1) 95% memory compression via 5-trits-per-byte Base-243 packing, (2) deterministic Hestenes Jacobi SVD basis extraction via total_cmp, (3) closed-form ALS leaf polish monotonically dampening quantization shock, and (4) SO(5) double-plane rotation commutativity on peripheral leaves.

Physical Verification and Measured Invariants

Test / Invariant	Physical Target / Dimension	Measured Result
Full-Rank HOSVD Roundtrip	X in $R^{\wedge}(4 \times 4 \times 4)$, $R=(4,4,4)$	Bit-exact (error < 10^{-5})
3D Low-Rank S13 Fitting	X in $R^{\wedge}(8 \times 8 \times 8)$, $R=(3,3,3)$	error = 0.428, Compression 2.1x
5D Geodesic Sky Projection	X in $R^{\wedge}(5 \times 5 \times 5 \times 5 \times 5)$, $R=(2,2,2,2,2)$	Compression 7.8x, Delta E < 0
SO(5) Leaf Commutativity	Double Givens Plane (theta=0.3, phi=0.7)	diff < 10^{-5} (Exact Commutation)
Sentinel Trapping	Byte states 243..255	100% Rejection (SentinelDetected)